YOUSEF MOHAMED AHMED

Backend Developer | Node.js | NestJS | TypeScript | PostgreSQL | MongoDB
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PROFESSIONAL SUMMARY

Fourth-year Information Technology student at Zagazig University seeking a Backend Node.js internship or junior backend role. Experienced in building REST APIs using Node.js, TypeScript, NestJS, and Express.js with PostgreSQL, MongoDB, and Redis. Focused on concurrency-safe business logic, database integrity, secure authentication and authorization, BullMQ background jobs, and real-time updates.

TECHNICAL SKILLS

Languages & Frameworks: Node.js, TypeScript, JavaScript (ES6+), Express.js, NestJS (Modules, Dependency Injection, Guards, Pipes, Interceptors)

Databases & Caching: PostgreSQL (TypeORM, Migrations, Transactions, Locking), MongoDB (Mongoose, Transactions, TTL Indexes), Redis (Caching, Distributed Locks, Lua Scripts)

Auth & Security: JWT, Refresh Token Rotation & Reuse Detection, Google OAuth 2.0, Passport.js, bcrypt, RBAC, Resource-Level Authorization, OTP Verification, Helmet, CORS, Rate Limiting, Idempotency Keys, Zod, Joi

Queues & Real-time: BullMQ (Delayed Jobs, Workers, Retries), Socket.IO with Redis Adapter, WebSockets

APIs & Integrations: REST API Design, Swagger / OpenAPI, Paymob Payments, Cloudinary, Multer, Nodemailer

Tools & Logging: Git, GitHub, Postman, npm, Pino, Morgan, Winston

Frontend: HTML5, CSS3, Bootstrap

Languages: Arabic (Native), English (Intermediate, Technical)

PROJECTS

PropFlow — Property & Rental Management API | *NestJS, TypeScript, PostgreSQL, Redis*

- Built a REST API with four roles (tenant, owner, maintenance staff, admin) covering properties and units, rental requests, lease lifecycle, maintenance workflow, notifications, analytics and audit logging.
- Enforced data integrity in PostgreSQL with transactions, row-level locks, a date-range exclusion constraint against overlapping leases and a partial unique index allowing one pending request per tenant and unit.
- Implemented JWT authentication with refresh-token family rotation and reuse detection, httpOnly refresh cookie, Google OAuth and Redis-backed email OTPs; combined global role guards with resource-level ownership checks to prevent IDOR.
- Added Redis cache-aside with explicit invalidation, a custom Redis storage for @nestjs/throttler, magic-byte upload validation with Cloudinary, Zod-validated configuration and migration-only schema management.

Clinic Appointment Booking System | *TypeScript, Express, MongoDB, Redis, BullMQ*

- Serialized concurrent reservations with a Redis distributed lock (SET NX, Lua compare-and-delete release, retry with backoff), backed by a unique compound index on doctor, date and start time.
- Created pending appointments with a BullMQ delayed expiry job and a partial MongoDB TTL index as a safety net; notifications run through a separate queue with exponential-backoff retries.
- Integrated Paymob payments (iframe flow, HMAC-SHA512 callback verification, refunds) and live updates over Socket.IO with the Redis adapter; added JWT rotation with reuse detection, Google OAuth, Redis rate limiting and idempotency keys.

Event Booking & Queue Management System | *TypeScript, Express, MongoDB, Redis, BullMQ*

- Prevented overselling with an atomic Redis Lua capacity counter and MongoDB transactions; BullMQ delayed jobs expire unconfirmed bookings.
- Implemented a FIFO waitlist on Redis sorted sets with BullMQ workers for promotion, confirmation deadlines and notifications, plus real-time queue events over Socket.IO.
- Added Google OAuth, email OTP, refresh-token rotation with reuse detection, Lua rate limiting and idempotency keys.

FreelanceHub Platform API | *Express, MongoDB, Redis, BullMQ*

- Built a freelance marketplace API (JavaScript) for jobs, proposals, contracts and notifications with client, freelancer and admin roles; BullMQ email worker, Redis response caching, Google OAuth and email OTP.
- Prevented double acceptance by moving a job from open to in-progress with a conditional findOneAndUpdate before creating the contract and rejecting competing proposals.

EDUCATION & TRAINING

Bachelor of Science in Information Technology — Faculty of Computers and Information, Zagazig University, Egypt Expected 2027

MEAN Stack Development (210 hours) — National Telecommunication Institute (NTI)

The Complete JavaScript Course 2025 (71 hours) — Udemy

Modern JavaScript & Backend Node.js (40 hours) — Manara